SC4063_NETWORK_SECURITY

COMPLETE

SYS_START: 11/04/2026 12:46:15 LAST_SYNC: 11/04/2026 16:48:14

Agentic Network Forensic Report – SC4063 Network Security

Executive Summary

This report was generated by a constrained forensic agent that uses alert-driven evidence extraction and a bounded reasoning loop. Its findings should be reviewed by a human analyst before external distribution.

Scope and Assumptions

SC4063 Network Security investigation: a ransomware incident attributed to the Lynx group.

The available evidence is a 9-day PCAP captured from a core-switch tap. There is no

endpoint telemetry or memory image, so all conclusions must be grounded in network data.

Evidence Sources

- Alert JSON: `/Users/aaron/Desktop/Network_Security_Project/network/34936-sensor-alert-1a30b94e-4fe6-4f94-9d7e-c289ba1c7dcc.json`

- Zeek JSON: `/Users/aaron/Desktop/Network_Security_Project/network/34936-sensor-zeek-b0032190-fe13-4910-aae9-2907793ab566.json`
- PCAP dir: `/Users/aaron/Desktop/Network_Security_Project/network/pcap`

Ingest Statistics

METRIC	VALUE
Total Suricata alerts	161,663
Alerts – c2	112,500
Alerts – trojan	26,032
Alerts – other	16,198
Alerts – policy	6,559
Alerts – lateral	356
Alerts – scan	18
Zeek lines scanned	9,323,452
Zeek records matched	9,291,241
Total PCAPs indexed	117
Targeted PCAPs opened	117
DNS queries extracted	50000
TLS sessions extracted	46744
HTTP requests extracted	0
SMB sessions extracted	0
RDP sessions extracted	0

Top Alert Rules

RULE	COUNT
ThreatFox botnet C2 traffic (domain - confidence level: 100%)	111930
ET MALWARE Known Hostile Domain ilo.brenz .pl Lookup	24329
ET DYN_DNS DNS Query to DynDNS Domain *.ufcfan .org	10585
ETPRO INFO Observed DNS Query to Dynamic DNS Service	6557
ET MALWARE UNK_FlappyBird Domain in DNS Lookup (onnetmais .org)	1057
ET DNS Reply Sinkhole - sinkhole.cert.pl 148.81.111.111	893
ET USER_AGENTS Microsoft Device Metadata Retrieval Client User-Agent	755
ET DNS Query to a .tk domain - Likely Hostile	587
ETPRO POLICY Observed DNS Query to Known ScreenConnect/ConnectWise Remote Deskto	576
ET REMOTE_ACCESS Observed DNS Query to Known ScreenConnect/ConnectWise Remote De	576

Targeted PCAP Files

- `34936-sensor-250305-00002430_redacted.pcap`
- `34936-sensor-250308-00002461_redacted.pcap`
- `34936-sensor-250308-00002473_redacted.pcap`
- `34936-sensor-250309-00002481_redacted.pcap`
- `34936-sensor-250309-00002488_redacted.pcap`
- `34936-sensor-250305-00002429_redacted.pcap`
- `34936-sensor-250303-00002384_redacted.pcap`
- `34936-sensor-250304-00002412_redacted.pcap`
- `34936-sensor-250304-00002393_redacted.pcap`
- `34936-sensor-250309-00002490_redacted.pcap`
- `34936-sensor-250303-00002386_redacted.pcap`
- `34936-sensor-250306-00002439_redacted.pcap`

- `34936-sensor-250309-00002491_redacted.pcap`
- `34936-sensor-250304-00002418_redacted.pcap`
- `34936-sensor-250304-00002402_redacted.pcap`
- `34936-sensor-250308-00002455_redacted.pcap`
- `34936-sensor-250308-00002464_redacted.pcap`
- `34936-sensor-250309-00002493_redacted.pcap`
- `34936-sensor-250307-00002447_redacted.pcap`
- `34936-sensor-250308-00002463_redacted.pcap`
- ... plus 97 more

Findings

A. Initial Access

Status: confirmed **Confidence:** HIGH **MITRE:** T1133

The internal host 10.128.239.57 was identified as patient zero, targeted by 305 unique external sources with 278,102 inbound RDP connections. The first external RDP connection was from 194.0.234.17 at 2025-03-01T18:20:01.130Z, followed by immediate internal DCE/RPC activity, indicating compromise. Credential spraying was evident from multiple RDP cookies with administrative usernames. Domain controllers 10.128.239.23 and 10.128.239.29 were identified through Kerberos and LDAP connections. Post-compromise activities included executable staging and SMB connections to multiple hosts. Alerts showed inbound traffic from known malicious IPs.

Evidence:

- `2025-03-01T18:20:01.130Z` | RDP | `194.0.234.17` → `10.128.239.57` | First external RDP connection to patient zero, indicating initial access. | _Artifact: zeek_conn_

LIVE MONITOR | FINDINGS | REPORT | TIMELINE | AGENT LOG

- `2025-03-01T18:20:01.530Z` | RDP | `147.45.112.100` → `10.128.239.57` | RDP cookie 'WATER\Adm' indicating credential spraying attempt. | _Artifact: zeek_rdp_

- `2025-03-06T23:06:23.109Z` | SMB | `10.128.239.57` → `10.128.239.36` | Executable 'SETUP.EXE' staged on internal host, indicating preparation for further compromise. | _Artifact: zeek_smb_
- `2025-03-01T18:22:42.424Z` | Alert | `20.106.236.198` → `10.128.239.57` | Inbound traffic from IP with poor reputation, indicating malicious targeting. | _Artifact: alerts_

Limitations:

- No endpoint telemetry available, relying solely on network data.

Recommended next steps:

- Investigate potential data exfiltration paths.
- Enhance monitoring on identified domain controllers.
- Review RDP access policies and strengthen authentication mechanisms.

B. Lateral Movement and Discovery

Status: confirmed **Confidence:** HIGH **MITRE:** T1046, T1021.002

The investigation confirms that the internal host 10.128.239.57 was the primary pivot for lateral movement within the network. This host connected to 52 unique internal targets using ports associated with lateral movement (445, 135, 3389, 5985). It also performed extensive DCERPC operations, including 240,175 NetrLogonSamLogonEx operations, indicating domain authentication attempts. Additionally, 10.128.239.23 executed 28,932 DRSGetNCChanges operations, suggesting potential DCSync attacks. SMB access to IPC$ shares on domain controllers was observed, indicating attempts to establish administrative connections. RDP connections from 10.128.239.57 to 26 unique internal targets further confirm its role in lateral movement.

Evidence:

- `2025-03-01T18:20:01.130Z` | DCERPC | `10.128.239.57` → `10.128.239.98` | 10.128.239.57 performed 240,175 NetrLogonSamLogonEx operations, indicating extensive domain authentication attempts. | _Artifact: zeek_dce_rpc_
- `2025-03-01T18:20:01.130Z` | DCERPC | `10.128.239.23` → `10.128.239.98` | 10.128.239.23 executed 28,932 DRSGetNCChanges operations, suggesting potential DCSync attacks. | _Artifact: zeek_dce_rpc_
- `2025-03-01T18:20:01.130Z` | SMB | `10.128.239.163` → `10.128.239.23` |

10.128.239.163 accessed \jjjjjjjwtDC8.water.domain-ees3Ai.local\IPC$, indicating attempts to establish administrative connections. | _Artifact: zeek_smb_

- `2025-03-01T18:20:01.130Z` | RDP | `10.128.239.57` → `10.128.239.98` | 10.128.239.57 connected to 26 unique internal targets over RDP, confirming its role in lateral movement. | _Artifact: zeek_conn_

Limitations:

- The query for delete.me probing was blocked due to a keyword restriction, limiting insights into specific probing activities.

Recommended next steps:

- Investigate potential credential theft and account manipulation activities.
- Monitor for any further unauthorized access attempts from identified pivot hosts.

C. Exfiltration

Status: suspected **Confidence:** MEDIUM **MITRE:** T1567

The investigation identified potential exfiltration activities involving the internal IP 10.128.239.57 connecting to the file-sharing service temp.sh (IP 51.91.79.17) on 2025-03-01 and 2025-03-02. Pre-exfiltration staging was evident with multiple archive files accessed or created on 2025-03-06, suggesting data preparation for exfiltration. However, there is no direct evidence of data transfer to the external IP in the connection logs, and no alerts were triggered for these activities.

Evidence:

- `2025-03-01T18:21:38.162Z` | TLS | `10.128.239.57` → `51.91.79.17` | Connection to temp.sh indicating potential exfiltration service use. | _Artifact: pcap_tls_
- `2025-03-06T23:11:03.956Z` | SMB | `10.128.239.57` → `10.128.239.176` | Access to archive file arrestees.zip, indicating pre-exfiltration staging. | _Artifact: zeek_smb_

Limitations:

- No direct evidence of data transfer to the external IP 51.91.79.17 in connection logs.
- No alerts were triggered for exfiltration activities.

Recommended next steps:

- Investigate other potential exfiltration methods or destinations.
- Review endpoint logs if available for additional evidence of data transfer.

D. Payload Deployment

Status: confirmed **Confidence:** HIGH **MITRE:** T1021.001, T1021.002

The ransomware was deployed primarily via SMB and RDP from the internal host 10.128.239.57. This host exhibited significant RDP fan-out, connecting to 26 unique internal destinations during the late period. Additionally, it was involved in transferring multiple executable files over SMB to various internal hosts, as evidenced by alerts for SMB2 NT Create AndX Request For an Executable File. External IP 194.0.234.17, previously identified as the initial access point, reconnected to 10.128.239.57 via RDP, suggesting manual triggering of the deployment.

Evidence:

- `2025-03-02T00:00:01.130Z` | RDP | `10.128.239.57` → `10.128.239.36` | Internal host 10.128.239.57 connected to 26 unique internal destinations via RDP during the late period, indicating significant fan-out. | _Artifact: zeek_conn_
- `2025-03-06T23:06:23.109Z` | SMB | `10.128.239.57` → `10.128.239.36` | Executable file SETUP.EXE transferred from 10.128.239.57 to 10.128.239.36, indicating payload staging. | _Artifact: zeek_smb_
- `2025-03-06T23:12:37.873Z` | SMB | `10.128.239.57` → `10.128.239.34` | Multiple executable files transferred from 10.128.239.57 to 10.128.239.34, as indicated by SMB2 NT Create AndX Request alerts. | _Artifact: alerts_
- `2025-03-02T00:00:01.985Z` | RDP | `194.0.234.17` → `10.128.239.57` | External IP 194.0.234.17 reconnected to 10.128.239.57 via RDP, suggesting manual triggering of ransomware deployment. | _Artifact: zeek_conn_

Limitations:

- The exact payload content is unknown due to lack of endpoint telemetry.

Recommended next steps:

- Investigate potential persistence mechanisms used by the attacker.
- Review backup integrity and restoration procedures.

Timeline

- `2025-03-01T18:20:01.130Z` | Initial Access | `194.0.234.17` → `10.128.239.57` | First external RDP connection to patient zero, indicating initial access.
- `2025-03-01T18:20:01.130Z` | Lateral Movement and Discovery | `10.128.239.57` → `10.128.239.98` | 10.128.239.57 performed 240,175 NetrLogonSamLogonEx operations, indicating extensive domain authentication attempts.
- `2025-03-01T18:20:01.130Z` | Lateral Movement and Discovery | `10.128.239.23` → `10.128.239.98` | 10.128.239.23 executed 28,932 DRSGetNCChanges operations, suggesting potential DCSync attacks.
- `2025-03-01T18:20:01.130Z` | Lateral Movement and Discovery | `10.128.239.163` → `10.128.239.23` | 10.128.239.163 accessed \jjjjjjjwtDC8.water.domain-ees3Ai.local\IPC$, indicating attempts to establish administrative connections.
- `2025-03-01T18:20:01.130Z` | Lateral Movement and Discovery | `10.128.239.57` → `10.128.239.98` | 10.128.239.57 connected to 26 unique internal targets over RDP, confirming its role in lateral movement.
- `2025-03-01T18:20:01.315Z` | Initial Access | `10.128.239.57` → `10.128.239.29` | Immediate internal DCE/RPC activity following external RDP connection, indicating compromise.
- `2025-03-01T18:20:01.530Z` | Initial Access | `147.45.112.100` → `10.128.239.57` | RDP cookie 'WATER\Adm' indicating credential spraying attempt.
- `2025-03-01T18:21:38.162Z` | Exfiltration | `10.128.239.57` → `51.91.79.17` | Connection to temp.sh indicating potential exfiltration service use.
- `2025-03-01T18:22:42.424Z` | Initial Access | `20.106.236.198` → `10.128.239.57` | Inbound traffic from IP with poor reputation, indicating malicious targeting.
- `2025-03-02T00:00:01.130Z` | Payload Deployment | `10.128.239.57` → `10.128.239.36` | Internal host 10.128.239.57 connected to 26 unique internal destinations via RDP during the late period, indicating significant fan-out.

- `2025-03-02T00:00:01.985Z` | Payload Deployment | `194.0.234.17` → `10.128.239.57` | External IP 194.0.234.17 reconnected to 10.128.239.57 via RDP, suggesting manual triggering of ransomware deployment.
- `2025-03-06T23:06:23.109Z` | Initial Access | `10.128.239.57` → `10.128.239.36` | Executable 'SETUP.EXE' staged on internal host, indicating preparation for further compromise.
- `2025-03-06T23:06:23.109Z` | Payload Deployment | `10.128.239.57` → `10.128.239.36` | Executable file SETUP.EXE transferred from 10.128.239.57 to 10.128.239.36, indicating payload staging.
- `2025-03-06T23:11:03.956Z` | Exfiltration | `10.128.239.57` → `10.128.239.176` | Access to archive file arrestees.zip, indicating pre-exfiltration staging.
- `2025-03-06T23:12:37.873Z` | Payload Deployment | `10.128.239.57` → `10.128.239.34` | Multiple executable files transferred from 10.128.239.57 to 10.128.239.34, as indicated by SMB2 NT Create AndX Request alerts.

Methodology and Guardrails

The agent follows a four-phase alert-driven pipeline: (1) Suricata alert ingestion, (2) Zeek JSON correlation filtered to IOC IPs, (3) targeted PCAP selection by timestamp, (4) deep tshark analysis on selected PCAPs only.

- Read-only PCAP handling
- Fixed tool allowlist (initial_access, lateral_movement, exfiltration, payload_delivery)
- Evidence-linked conclusions – every finding cites actual log entries
- Confidence labels (HIGH/MEDIUM/LOW) with explicit limitations
- Safe abstention when evidence is weak
- Results cached in `ingest/` to avoid re-scanning the 30 GB Zeek file

Cost and Efficiency Notes

- The Zeek JSON is streamed once and matched records are cached; re-runs are fast.
- tshark is invoked only on the targeted PCAP subset, not all 129 files.

- Deterministic mode has zero model cost; LLM mode adds cost only for planning.
- This keeps the expensive reasoning layer narrow and auditable.